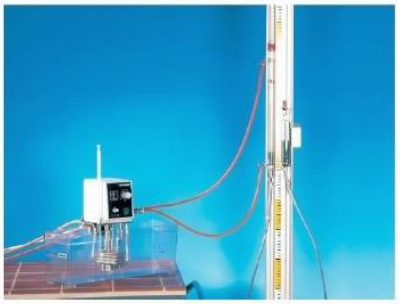
T. C.
İSTANBULÜNİVERSİTESİ
FEN FAKÜLTESİ
FİZİK BÖLÜMÜ

THERMODYNAMICS AND STATISTICAL PHYSICS LABORATORY



2026

T.C.

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THERMODYNAMICS AND STATISTICAL MECHANICS LABORATORY EXPERIMENTAL LITERATURE & MANUALS

[2026]

THERMODYNAMICS AND STATISTICAL MECHANICS LABORATORY

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Name-Surname:

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Experiments	Date	Sign
EQUATION OF STATE FOR IDEAL GASES		
HEAT CAPACITY OF METALS		
MECHANICAL EQUIVALENT OF HEAT		
THERMAL AND ELECTRICAL CONDUCTIVITY OF METALS		
HEAT CONDUCTION AND ISOLATION		
THERMAL EQUATION OF STATE AND CRITICAL POINT		
MAXWELL'S VELOCITY DISTRIBUTION		
STEFAN-BOLTZMANN LAW OF RADIATION		

DENEYLERİN İMZALARI ALINDIKTAN SONRA BU İMZA SAYFASI DÖNEM SONUNDA DERSİN HOCASINA TESLİM EDİLECEKTİR.

Date: …../…../…….

Asisst. Prof. Dr. Deniz BOZOĞLU PARTO

LABORATORY RULES

1. Arriving at the laboratory at the specified times is an indispensable element of work discipline. Therefore, please arrive at the laboratory at the announced time.
2. Before coming to the laboratory, be sure to obtain prior information about the experiment you will be conducting that day.
3. You must bring the necessary materials for your experiment, such as a ruler, calculator, and millimeter paper, which will not be provided in the laboratory.
4. Do not speak loudly in order to avoid disrupting the working environment in the laboratory.
5. Eating and drinking are prohibited in the laboratory.
6. Remember to turn off your mobile phones from the moment you enter the laboratory.
7. From the moment you enter the laboratory, do not touch the experiment sets without the permission and instructions of the instructor.
8. Each student should work at their own desk and should not interfere with the experiments conducted by others at other desks or with the experiment sets.
9. Equipment should be used carefully. The student is responsible for replacing any materials damaged due to carelessness or lack of knowledge.
10. The experiment table, tools, and equipment must be kept clean and tidy. After the experiment is finished, the experiment table should be cleaned and the tools and equipment put back in their place.
11. The experiment sets are set up in a way that will not endanger your safety. However, in case of an accident, immediately inform the Instructor.
12. If there is any problem with the experiment sets, immediately inform the Instructor.

PREFACE

The first section of the Thermodynamics and Statistical Mechanics Laboratory Experiment Booklet provides essential statistical information for the experiments. The second section includes details for conducting experimental studies on the following topics relevant to this course: the Equation of State for an Ideal Gas, Heat Capacity of Metals, the Mechanical Equivalent of Heat, Thermal and Electrical Conductivity of Metals, Heat Conduction and Insulation, Thermal Equation of State and Critical Point, Maxwell's Velocity Distribution, and the Stefan-Boltzmann Radiation Law.

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8. **STEFAN-BOLTZMANN LAW OF RADIATION**……

01- EQUATION OF STATE FOR IDEAL GAS

PRINCIPLE

The state of a gas is determined by temperature, pressure and amount of substance. In the limiting case of ideal gases, these state variables are related by the ideal gas law. For a change of state under isobaric conditions, this equation becomes Gay-Lussac's first law, while under isochoric conditions it becomes Amonton's law and for isothermal processes it becomes Boyle and Mariotte's law.

TASKS

1. Analyse the validity of the three gas laws for a constant quantity of gas.
2. Calculate the universal gas constant, the coefficient of thermal expansion, and the coefficient of thermal stress using the relationships obtained.

THEORY

The ideal gas law is illustrated in this experiment with the help of three experiments. The aim is to show that this law is a good approximation for the behaviour of many gases under many conditions, despite numerous limitations.

Boyle-Mariotte's law reads: $\boldsymbol{p.V = constant}$ **resp.** $\boldsymbol{p \sim 1/V}$ and states that the pressure of an ideal gas at constant temperature is inversely proportional to the volume. This means that if the pressure of $\boldsymbol{p_1}$ on $\boldsymbol{p_2}$ increases, the volume of $\boldsymbol{V_1}$ on $\boldsymbol{V_2}$ and vice versa:

$$\frac{p_1}{p_2} = \frac{V_1}{V_2} \tag{1}$$

This follows from the formula for the ideal gas law, since in the experiment the temperature T and the amount of substance n remain constant:

$$p.V = n.R.T \quad \text{resp.} \quad p = \frac{n.R.T}{V} \tag{2}$$

This means that the volume of a gas in a gas cylinder under high pressure is less than under normal pressure.

The law of Gay-Lussac is: $\boldsymbol{V/_T\, constant}$ **resp.** $\boldsymbol{V \sim T}$ and states that the volume of an ideal gas at constant pressure is proportional to the temperature. This means that if the temperature of $\boldsymbol{T_1}$ on $\boldsymbol{T_2}$ increases, the volume of $\boldsymbol{V_1}$ on $\boldsymbol{V_2}$ and vice versa:

$$\frac{\boldsymbol{V_1}}{\boldsymbol{V_2}} = \frac{\boldsymbol{T_1}}{\boldsymbol{T_2}} \tag{3}$$

This follows from the formula for the ideal gas law, since in the experiment the pressure and the amount of substance remain constant:

$$\boldsymbol{p.V = n.R.T} \quad \text{resp.} \quad \boldsymbol{V = \frac{n.R.T}{p}} \tag{4}$$

This law is used in pressure vessels, air conditioning systems, balloons and in the analysis of chemical reactions and meteorological conditions.

The law of Amontons reads: $\boldsymbol{p/_T\, constant}$ **resp.** $\boldsymbol{p \sim T}$ and states that the pressure of an ideal gas at constant volume is proportional to the temperature. This means that if the temperature of $\boldsymbol{T_1}$ on $\boldsymbol{T_2}$ increases, the pressure from $\boldsymbol{p_1}$ on $\boldsymbol{p}$ and vice versa:

$$\frac{\boldsymbol{p_1}}{\boldsymbol{p_2}} = \frac{\boldsymbol{T_1}}{\boldsymbol{T_2}} \tag{5}$$

This follows from the formula for the ideal gas law, since in the experiment the volume and the amount of substance remain constant:

$$\boldsymbol{p.V = n.R.T} \quad \text{resp.} \quad \boldsymbol{p = \frac{n.R.T}{V}} \tag{6}$$

In contrast to Gay-Lussac's law, Amonton's law states that at constant volume the *Pressure* of a gas increases when the *Temperature* is increased.

The ideal gas law sets the variables pressure p, temperature T, volume V, and the amount of substance n are related to each other. It is defined as

$$\boldsymbol{p.V = n.R.T} \tag{7}$$

with the Avogadro or ideal gas constant R.

The experimentally obtained diagrams show that there is a linear relationship between the respective external state variables of pressure, temperature and volume analogous to the three gas laws.

The state of a gas is given by the state variables temperature T, pressure p, volume V and the amount of substance n, which interactively determine one another. Thus, the dependence of the volume on the variables temperature and pressure at a constant amount of substance ($n = const.$; $dn = 0$) enclosed amount of gas in the measuring tube) is given by the total differential.

$$dV = \left(\frac{\partial V}{\partial T}\right)_{p,n} dT + \left(\frac{\partial V}{\partial p}\right)_{T,n} dV + \left(\frac{\partial p}{\partial n}\right)_{T,V} dn \tag{8}$$

Analogously, the following is valid for the change of pressure with T and V

$$dp = (\frac{\partial p}{\partial T})_{V,n}\ dT\ +\ (\frac{\partial p}{\partial V})_{T,n} dp + (\frac{\partial V}{\partial n})_{T,V} dn \tag{9}$$

In the evaluation, we will concentrate on the colour-coded terms relevant to the three gas laws.

The relationship (10) is simplified for a certain amount of substance ($n = const.\ ;\ \ dn = 0$ enclosed gas volume in the gas syringe) and an isothermal change of state ($T = const.; dT = 0$) to:

$$dp = (\frac{\partial p}{\partial T})_{V,n}\ dT \tag{10}$$

The partial differential quotient $({}^{\partial p}/_{\partial \boldsymbol{V}})_{T,n} dV$ corresponds geometrically to the slope of a tangent to the function ***V=f(p)*** and thus characterises the interdependence of pressure and volume. The degree of this dependency is characterised by the output pressure p_0 determined. Therefore, one defines the following intensive variables by referring them to V or p or V_0 or p_0 at T_0= 273.15 K:

$$\gamma_0 = \frac{1}{V_0}(\frac{\partial p}{\partial T})_{p,n} \tag{11}$$

$$\beta_0 = \frac{1}{p_0}(\frac{\partial p}{\partial T})_{V,n} \tag{12}$$

$$\chi_0 = -\frac{1}{V_0}(\frac{\partial V}{\partial p})_{T,n} \tag{13}$$

γ_0: coefficient of thermal expansion

β_0: coefficient of thermal tension

χ_0: coefficient of cubic compressibility

For the limiting case of an ideal gas (pressure which is sufficiently small, temperature which is sufficiently high), these relationships can be derived from empirically determined laws, which each describe the correlation between two state variables

For isobaric process control ($p = const.; dp = 0$) eqn. (8) becomes

$$dV = V_0 \gamma_0 dT \tag{14}$$

For $\gamma_0 = const.$, the integration of this differential equation results in

$$V = V_0\left(1 + \gamma_0(T - T_0)\right) = V_0(1 + \gamma_0 \vartheta) \tag{15}$$

ϑ: Temperature on Celcius scale

or

$$\frac{\boldsymbol{V_0}}{\boldsymbol{T_0}} = \frac{\boldsymbol{V}}{\boldsymbol{T}} \tag{16}$$

or

$$V = const.T \tag{17}$$

In accordance with this correlation, which was discovered by Gay-Lussac, the graphic presentation of the volume as a function of the temperature supplies rising straight lines (Fig. 2) with V = 0 for T = 0.

At isochoric conditions (V = const, dV = 0), the following results analogously from (9)

$$dp = p_0 \beta_0 dT \tag{18}$$

For $\beta_0 = const.$, the integration of this differential equation results in

$$p = p_0 \left(1 + \beta_0 (T - T_0)\right) = p_0 (1 + \beta_0 \vartheta) \tag{19}$$

and

$$\frac{\boldsymbol{p_0}}{\boldsymbol{T_0}} = \frac{\boldsymbol{p}}{\boldsymbol{T}} \tag{20}$$

or

$$p = const.T \tag{21}$$

These equations are the expression of Charles' (Amontons') law and describe the linear increase in the pressure with increasing temperature (Fig. 2).

For the isothermal expansion and compression (T = const., dT = 0) of a constant quantity of gas, the following is valid:

$$dV = V_0 \chi_0 dp \tag{22}$$

or

$$V_0 p_0 = V p_0 \tag{23}$$

and

$$V = const.\frac{\mathbf{1}}{\boldsymbol{p}} \tag{24}$$

The graphic presentation of this correlation, which was investi gated by Boyle and Mariotte, results in hyperbolas for V = f(p) (Fig. 3); in contrast, for V = f(1/p) straight lines are obtained (Fig. 4).

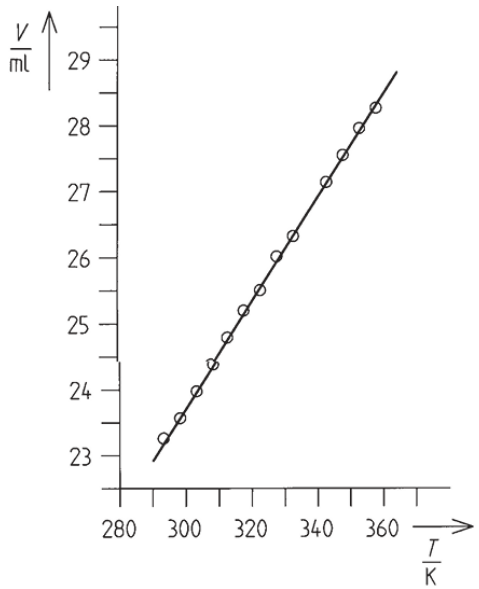


Fig. 2: Dependence ot the volume V on the temperature T at constant pressure (p = 100.6 kPa) and a constant substance quantity (n = 0.9536 mmol).

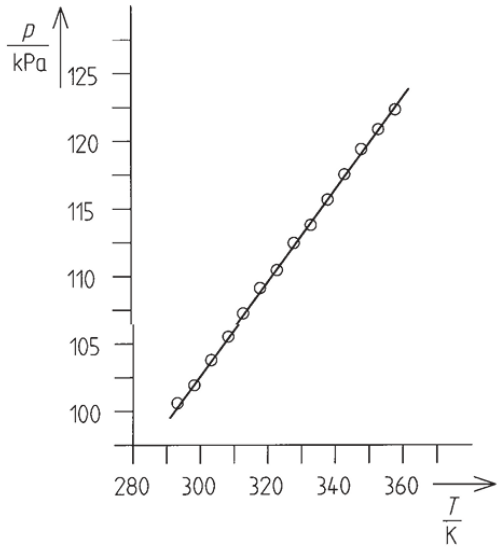


Fig. 3: Pressure p as a function of the temperature T at constant volume (V = 2.326 · 10^{-5} m^3) and a constant substance quantity (n = 0.9536 mmol).

Fig. 4: Correlation between pressure p and volume V for a constant quantity of air (n = 0.9536 mmol) during an isothermic change of state (T = 298.15 K).

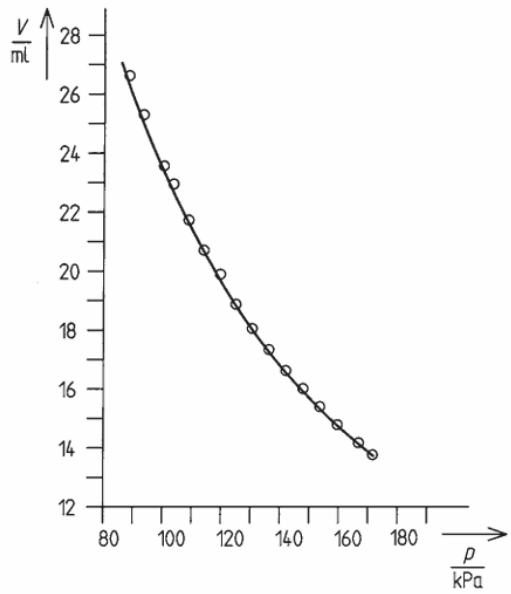
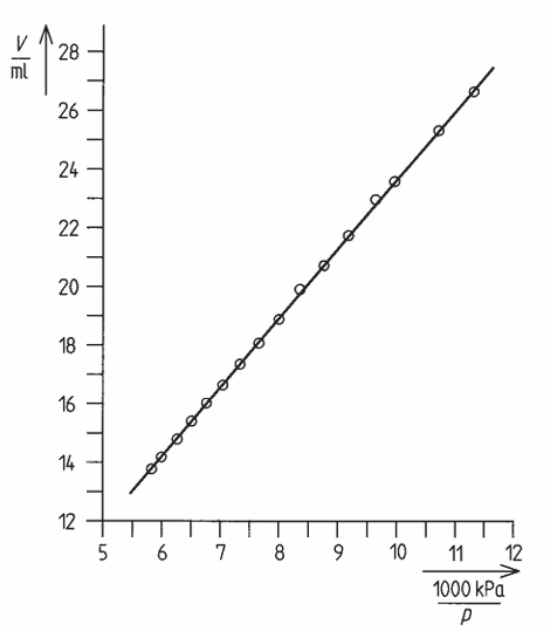


Fig. 5: Volume V as a function of the reciprocal pressure $1/p$ at constant temperature (T= 298.15 K) and a constant substance quantity (n = 0.9536 mmol).



By combining the equations (16) or (17) with (7.1) one direct ly obtains via

$$\frac{p_0 V_O}{T_0} = \frac{p_1 V_1}{T_1} = \frac{pV}{T} \tag{25}$$

the general or thermal equation of state for ideal gases (9) with the universal gas constant R.

$$\boldsymbol{p.V = n.R.T} \tag{26}$$

This equation allows retroactive comprehension of the correlations obtained for equations (17), (21) and (24). Thus from the slopes of these linear relationships.

$$\left(\frac{\partial V}{\partial T}\right)_{p,n} = V_0 \gamma_0 = \frac{nR}{p} \tag{27}$$

$$\left(\frac{\partial p}{\partial T}\right)_{V,n} = V_0 \beta_0 = \frac{nR}{V} \tag{28}$$

$$\left(\frac{\partial V}{\partial p^{-1}}\right)_{T,n} = nRT \tag{29}$$

which can be obtained experimentally, the gas constant R can be calculated if the enclosed quantity of gas n is known. This is equal to the quotient (30) of the volume V and the molar volume V_m.

$$n = \frac{V}{V_m} \tag{30}$$

At T_0 = 273.15 K and p_0 = 101.325 kPa (standard conditions) this amounts to V_0= 0.022414 $m^3 \cdot mol^{-1}$. A volume V measured at p and T must first be reduced to these conditions using Eq. (25).

Moreover, from the equations (27) and (28) the coefficients γ_0 and β_0 can be determined. The required values for the volume V_0 and the pressure p_0 can be obtained by extrapolation to T_0 = 273.15 K on the corresponding graphs or from the corresponding regression equations or calculated using the rela tionships (16) and (20). Therefore, it is possible to calculate the compressibility coefficient x_0 from the total differential (8) using

$$x_0 = \frac{1}{p_0}\frac{y_0}{\beta_0} = \frac{1}{101.325kPa} \cdot \frac{\gamma_0}{\beta_0} \tag{31}$$

The values for γ_0 and β_0 which are now known.

Experiments with a constant quantity of gas (n = 0.9536 mmol) calculated according to the relations (25) and (30) confirm the validity of the investigated gas laws with their resulting linear correlations – which are in accordance with the equations (17), (21) and (24).

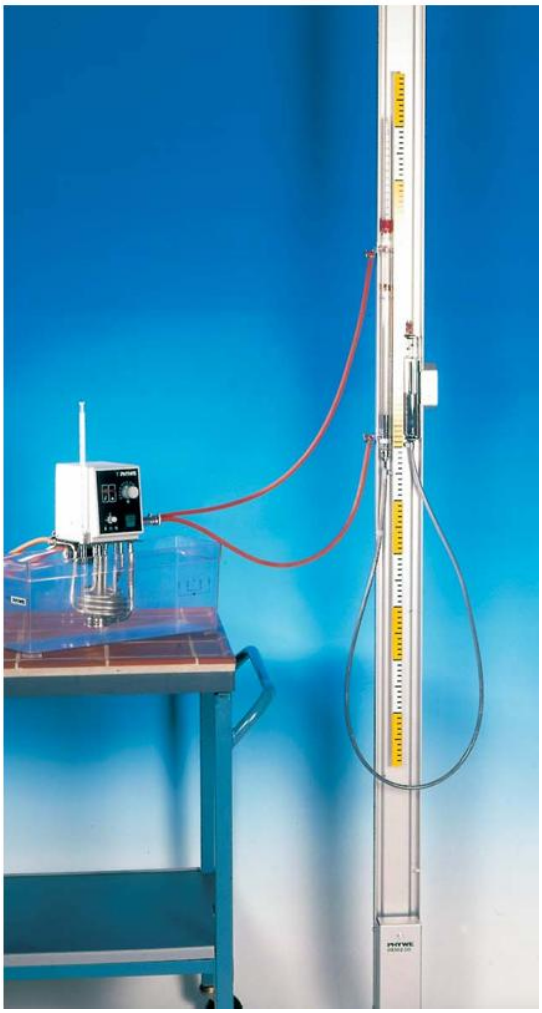
Equipment

Gas law units, thermostat, thermometer, thermostat bath.

Set-up and Procedure

The experiment is set-up according to Fig. 1 and connected with the pump branch unit of the thermostat (Secure the tubing connections with hose clips!). Fill the reservoir on the demonstration device carefully with mercury according to the operating instructions until the measuring tube is filled up to about one-quarter of the gra duated measuring range with mercury (equal levels of mercury in the measuring tube and the levelling container). Fill the bath of the circulating thermostat with distilled or de mineralised water to avoid furring. Connect the cooling coil of the thermostat to the water supply line using rubber tubing

Fig. 1: Experimental set-up: Equation of state of ideal gases



Part I

1. Carefully fill the thermostat bath with water.
2. Set the thermostat to 30°C. Wait until the temperature stabilizes.
3. By moving the gas laws unit, examine the pressure-volume relationship at constant temperature and record it in Table I.
4. Draw the P-V graph. Using the ideal gas equation, calculate the universal gas constant (R) for each measurement. Calculate the mean value (R_{ort}) and the standard deviation (ΔR). Record the values you found in Table I.

Table I

T = sabit= ..... ^{0}C= ……K					
	L(cm)	Δh(cm)	V(ml)	p(kPa)	R(J/mol.K)
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
R= R_{ort} ±ΔR=					

Part II

1. Equalize the mercury columns by moving the gas laws unit at 30°C. Measure the length of the air column L0 corresponding to this situation, calculate the corresponding V0 = constant value, and record it in the tables.
2. Bring the temperature to 40°C. Measure the length of the air column in the measuring tube by setting Δh=0, calculate the corresponding volume value, and record it in Table II. Bring the mercury level in the measuring tube to the L0 position. Read the corresponding εh value, calculate the corresponding pressure value, and record it in Table III.
3. Repeat the same steps, increasing the temperature by 10°C at a time, and take measurements up to 60°C.

4. Using Table II; P=Constant, draw the V-T graph. Calculate the thermal expansion coefficient using equation

(8).

5. Using Table III; Draw the P-T graph where V=Constant. Calculate the thermal stress coefficient using equation **(9).**

Tablo II.

p=sabit=…….....kPa			
t(^{0}C)	L(cm)	V(ml)	T(K)
30			
40			
50			
60			

Tablo III.

V =sabit=… ..... ml			
t(^{0}C)	Δh(cm)	P(kPa)	T(K)
30			
40			
50			
60			

QUESTIONS RELATED TO THE EXPERIMENT

1. The volume of substances in the gaseous state is greater than in their solid and liquid states. Explain why.
2. Are there quantum correlations between the particles that make up an ideal gas? Explain.
3. Which statistic do you think the distribution of an ideal gas follows?
4. What is Avogadro's principle? Research it.
5. Obtain the equation of state for an ideal gas.
6. Derive equation (4).
7. What are the values of the universal gas constant in different units?
8. What is the value of the Boltzmann constant k?
9. What does it mean for a function to be an exact differential equation?
10. Derive equation (17) $\chi 0 = f(\alpha 0, \beta 0)$ from the exact differential equation (6). 11. Why are the expansion and compression coefficients not equal for gases?
11. Which of the physical quantities V, p, T, ε0, ε0, ε0 are intensive and which are extensive?
12. Which of the physical quantities used in this experiment are intensive and which are extensive?
13. What do isothermal expansion and compression mean? Explain.
14. What do isobaric expansion and compression mean? Explain.
15. Does the p-V diagram given in Figure 3 represent the state of real gases?
16. How can we tell that the gas has not liquefied by looking at the diagram given in Figure 3?

17. Is the equation of state given for an ideal gas valid for real gases?
18. Interpret the graph you drew. Are the results as you expected?

REFERENCES

1. PHYWE series of publication, Laboratory Experiments

02- HEAT CAPACITY OF METALS

PRINCIPLE

Heated specimens are placed in a calorimeter filled with water at low temperature. The heat capacity of the specimen is determined from the rise in the temperature of the water.

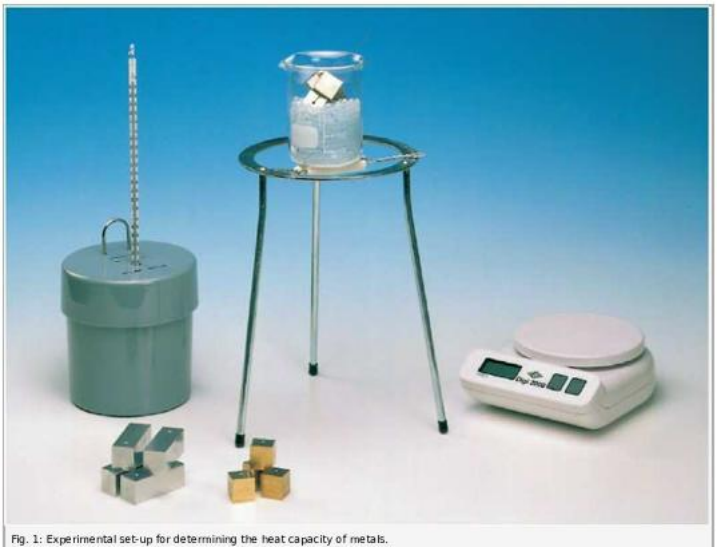


Fig. 1: Experimental set-up for determining the heat capacity of metals.

TASKS

1. To determine the heat capacity of the calorimeter by filling it with hot water and determining the rise in temperature.
2. To determine the specific heat capacity of aluminium, iron and brass.
3. To verify Dulong Petit's law with the results of these experiments.

THEORY

The heat capacity C of a substance is defined as the quotient of the quantity of heat absorbed δQ and the change in the temperature dT:

$$C = \frac{\delta Q}{dT} \tag{1}$$

and is the proportional to the mass the heated substance.

$$c = \frac{C}{m} \tag{2}$$

is the specific heat capacity.

The quantity of heat absorbed δQ depends on the conditions prevalling as the temperature rises, and a differentiation is made in particular between heat capacity C_V at constant volume V and heat capacity C_P at constant pressure p.

In accordance with the First Law of Thermodynamics (U=Internal Energy).

$$\delta Q = dU + pdV \tag{3}$$

is always greater than. In the case of solids, the change in volume is so small that we can write

$$C_P \cong C_V$$

Can abe calculated from the change in internal energy with temperature in accordance with (1) and (3):

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V \tag{4}$$

The internal energy U in a solid is essentially the result of lattice vibrations caused by heat.

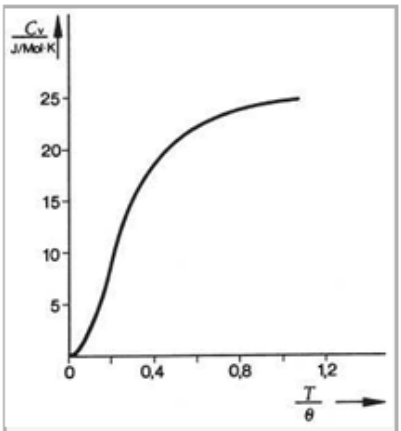


Fig. 2: Molar heat capacity of a solid in accordance with Debye's approximation.

According to Debye's theory, which considers lattice vibrations up to a limiting frequency, the heat capacity is given by

$$C_V(T) = 3N_k\left(\frac{T}{\theta}\right).3\int_0^{\theta/T}\left(\frac{z^4e^z d_z}{(e^z-1)^2}\right) = 3Nk.D\left(\frac{T}{\theta}\right) \quad (5)$$

where

$z = {}^{h}/_{kT}$,

$0 = {}^{hvD}/_{k}$, called the Debye temperature

h= Planck'S constant,

k= Boltzmann constant,

N= number of atoms in the volume considered.

$D\left(\frac{T}{\theta}\right)$ is called the Debye function.

For large values of $\left(\frac{T}{\theta}\right)$ the upper integration limit is small, the he integrand can be expanded and we obtain the law of Dulong-Petit:

$$C_V(T) = 3N_k$$

We thus obtain the molar heat capacity

$$C_m = 3N_L.k = 3R = 24.94\frac{J}{K} \quad (6)$$

where N_L is the Loschmidt number and R the gas constant.

Debye temperature:
Aluminium: 419 K

Copper: 335 K

Iron: 462 K

Zinc: 100 K

For the evaluation, the heat capacity is assumed to be constant in the temperature range considered.

The mixture (heat sharing) temperature ϑ_m is determined by extreapolating the curves plotted, as the sketch in Fig.3 shows.

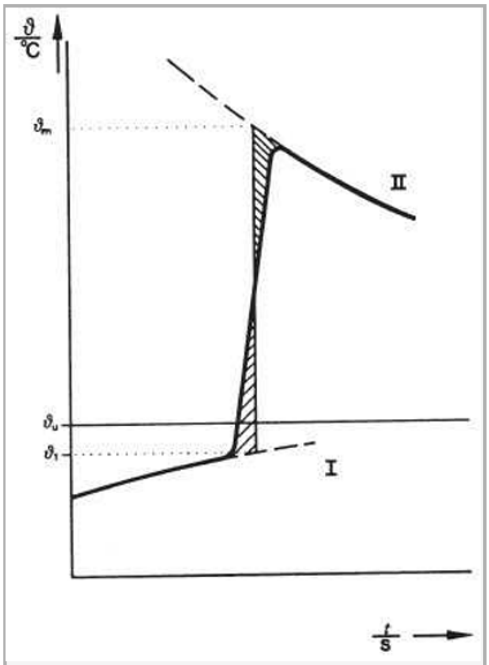


Fig. 3: Determination of the initial temperature ϑ_1 and the mixture temperature ϑ_m in the method of mixtures experimentwhen heat loss takes place. ϑ_u = temperature of the surrounding atmosphere.

The temperature before and after heat sharing are not constant because of the Exchange of heat with the surroundings; they approach the ambient temperature exponentially.

Fort he evaluation we draw a straight line parallel to the temperature axis in the graph of the temperature rise, so that the shaded parts in Fig.3 are euqal in area.

ϑ_m and ϑ_1 are read off at athe points where the straight line intersects the extended partial curves I and II.

1. First of all, the heat capacity of the calorimeter is determined:

$$C_k = c_w . m_w \frac{\vartheta_w - \vartheta_m}{\vartheta_w - \vartheta_k} \tag{7}$$

where ϑ_k is the temperature of the calorimeter before experiment, ϑ_m is the temperature of the mixture, ϑ_w is the temperature of the hot water, m_w is the mass of the water and c_w= 4.187 J/gk, the specific heat capacity of water.

The average value from several measurements (with the vessel half full) is

$$C_k = (66 \pm 1)\, J/K$$

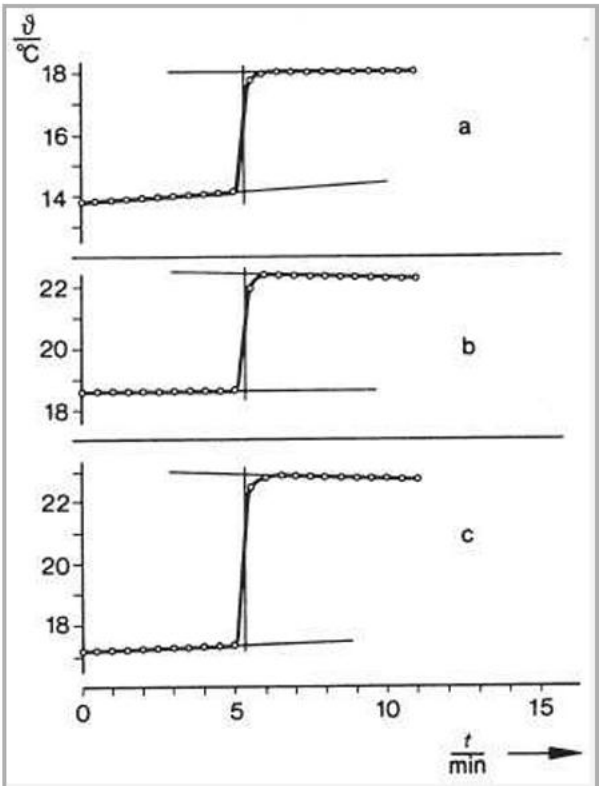


Fig. 4: Temperature as a function of time in the method of mixtures experiment a) steel, b) brass, c) aluminium.

2. The test pieces of mass $m_p$are heated in the boiling water. The boiling point ϑ_2 is determined as a function of the atmospheric pressure p:

$$\vartheta_2 = 100 + 0.0276(p - 1013hPa) - 0.000017(p - 1013hPa) \quad (8)$$

where p is in hPa and ϑ_2 in °C.

The specific heat capacity of the material from which the test pieces are made is obtained from the energy balance as:

$$c = \frac{c_w . m_w + C_k\,(\vartheta_m - \vartheta_1)}{m_p\,(\vartheta_2 - \vartheta_m)} \quad (9)$$

The values of molar heat capacity of brass and aluminium as measured in the experiment agree well with the values from Dulong and Petit's law.

The specific heat capacity of steel depends very much on the composition of the steel.

	$\frac{\vartheta_2}{°C}$	$\frac{m_p}{g}$	$\frac{m_w}{g}$	$\frac{c}{J/g{\cdot}K}$	$\frac{C_m}{J/Mol{\cdot}K}$
Brass	99.8	120.8	214.6	0.387	24.9
Aluminium	99.3	90.6	251.1	0.897	24.2
Steel	99.3	117.2	257.4	0.469	

SUMMARY

Specific Heat

Specific heat (also known as heat capacity) is the amount of heat required to change the temperature of 1 kilogram (1000 grams) of a substance by 1 °C, whether by adding heat or removing heat. To determine a substance's specific heat, heat must be added or removed. One effective method for this measurement is to bring the substance into contact with another substance at a different temperature, allowing them to reach thermal equilibrium. This approach is referred to as the "***mixing method."***

There are two primary rules, known as ***calorimetric laws***, that are applied when solving problems related to heat calculation and temperature change:

a) When substances with different temperatures are in contact, the hotter substances will cool down, while the colder substances will warm up until thermal equilibrium is achieved. In summary:

Heat Absorbed = Heat Given

b) The amount of heat required for an event to occur is equal to the amount of heat released when the opposite of that event occurs. The formula for the amount of heat needed to raise the temperature of an object with mass m from temperature t_1 to $t_2$is expressed as:

$$Q = m \cdot c(t_2 - t_1) = m \cdot c \cdot \Delta t$$

Atomic Heat

Atomic heat refers to the amount of heat needed to raise the temperature of one gram of an atom of a substance by 1 °C, which is denoted by C.

$$C = A \cdot c$$

where A is the atom weight.

Dulong-Petit Law

According to Dulong-Petit law, the atomic heat capacities of metals are approximately 6.4. This law allows for the determination of the heat capacities of all metals with known atomic weights.

Before determining the heat capacities of various materials, it is essential to first

establish the heat capacity of the calorimeter used in the experiment.

Table I:

ϑ_k(°C)	ϑ_w(°C)	m_w	ϑ_m (°C)	C_k(J/°K)

Equipment

Calorimeter container, metal parts, thermometer, heating tube, beaker, scale, fishing line, K-type thermocouple, multimeter.

Set-up and Prodecure:

Part I

1. Measure the calorimeter temperature before the experiment.
2. Pour (m_w) 300ml of water with hot water (approx. ϑ_w≈50°C) into the calorimeter.
3. Repeat the experiment to increase the accuracy of the measurement. Keep ready a vessel containing water at room temperature, so that the calorimeter can quickly be returned to room temperature after the measurement. Leave this water in the calorimeter for approx. 5 minutes and then empty and dry it.
4. Start the next measurement after another five minutes.
5. Record the values you obtained in Table I.

$$C_{k_{avg}} = \qquad j/K$$

Part II

1. Weigh each metals pieces to obtain their mass m_p.
2. Put a layer of glass beads about 3 cm deep in the bottom of the beaker so that the metal specimens do not touch it when they are being heated.
3. Fill the calorimeter with cold water of a known mass.
4. Leave the specimens in the boiling water for about 10 minutes, then drain them quickly and put them into the calorimeter.

5. Find the specific heat capacities for brass, steel, and aluminum and record the results in Table II.

Table II:

	ϑ_k(°C)	ϑ_2(°C)	ϑ_m(°C)	$m_p(gr)$	$m_w(gr)$	$c(J/gr.°K)$
Brass						
Aluminium						
Iron						

EXPERIMENT-RELATED QUESTIONS

1. What are heat and temperature? How can we distinguish these two physical quantities from each other?

2. Debye's theory can accurately explain heat capacity at low temperatures. Is Debye's theory valid at high temperatures? Research this.

3. Another theory proposed to explain heat capacity in solids is known as Einstein's theory. Re-establish heat capacity using Einstein's theory. Draw the heat capacity expression you obtained as a function of temperature and compare it with the results of Debye's theory.

4. What is latent heat? Is there a relationship between latent heat and heat capacity? Research this.

5. In which of the reversible and irreversible processes is the heat change greater?

6. What are the laws of thermodynamics? Explain them.

7. What physical consequences would you expect if a material had no heat capacity?

8. How can we increase the heat capacity of materials, mixtures, liquids, etc.? Have you ever wondered?

9. What is a solar pool? Research it.

10. Can a material with increased heat capacity be used in industry, manufacturing, and for the benefit of society?

11. Interpret the graph you drew. Are the results as you expected?

REFERENCES

1. PHYWE series of publications, Laboratory Experiments

03- MECHANİCAL EQUIVALENT OF HEAT

PRINCIPLE

In this experiment, a metal test body is rotated and heated by the friction due to a tensed band of synthetic material. The mechanical equivalent of heat for problem 1 is determined from the defined mechanical work and from the thermal energy increase deduced from the increase of temperature. Assuming the equivalence of mechanical work and heat, the specific thermal capacity of aluminium and brass is determined.

TASKS

1. Determination of the mechanical equivalent of heat.
2. Determination of the specific thermal capacity of aluminium and brass.

Fig.1: Experimental set-up

THEORY

For a long time it was debated as to whether the heat of a system, which used to be defined in terms of its temperature, was a form of energy or an independent magnitude which is conserved. During the first half of the nineteenth century, it was proven that mechanical energy due to friction is completely converted to heat, independently of the course of the transformation process and of the physical and chemical characteristics of the material used. Heat was accordingly defined as the energy of disorganized, macroscopically invisible molecular movements.

The quotient between realized mechanical work W and the quantity of heat Q generated through friction is called the mechanical equivalent of heat. In this experiment, mechanical work is performed by rotating the friction cylinder against the synthetic friction band's sliding frictional force. As weight M (cf. Fig. 2) is not accelerated when the crank is rotated, weight $F_G = M_g = 49\ N$ (M: mass of the weight, g: terrestrial gravitational acceleration) on one side and the sliding frictional force acting together with force on the suspension of the dynamometer on the other side, must cancel each other:

$$F_R = F_G - F_D \tag{1}$$

Friction work W is thus determined from the friction path of (r: radius of the cylinder, n: number of turns) and is found to be:

$$W = 2\pi r n F_R = 2\pi r n F_G - F_D \qquad (2)$$

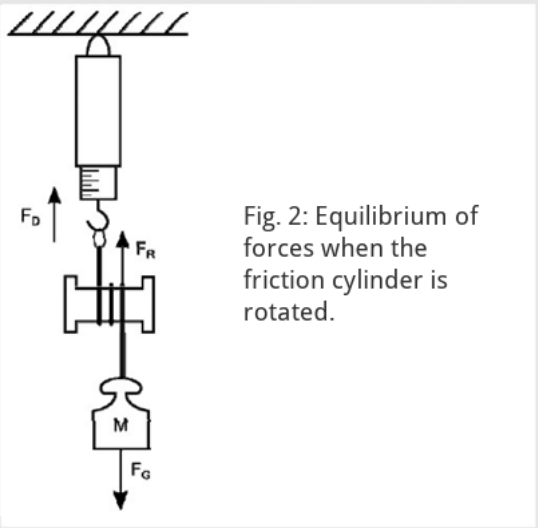


Fig. 2: Equilibrium of forces when the friction cylinder is rotated.

Whilst the synthetic band slides over the cylinder, a large portion of mechanical work is converted to thermal energy ΔQ through inelastic deformations of the friction material. Accordingly, the temperature T of the friction cylinder increases by a temperature difference ΔT:

$$\Delta Q = C_{tot} . \Delta T \qquad (3)$$

C_{tot} is the total thermal capacity of the heated parts. It is the sum of the thermal capacity C_{cyt} of the friction cylinder (m: mass, specific thermal capacity (CuZn: c = 0.385 J / gK)):

$$C_{cyl} = c . m \qquad (4)$$

Of the thermal capacity of the friction band ($C_{band} = 4\, J/K$) and of the thermal capacity of the thermometer ($C_{th} = 4\, J/K$). If the mass of the friction cylinder is m = 640 g, total thermal capacity amounts to:

$$C_{tot} = C_{cyl} + C_{band} + C_{th} = 254 \frac{J}{K} \qquad (5)$$

The thermal capacities of the insulated bearing and of the heat conducting paste were neglected. Fig. 3 displays a temperature-time diagram for an example of measurement. The shape of the graph makes it clear, that thermal energy is continuously given off to the environment while the crank is being turned. Hence, temperature difference ΔT is determined as is generally done for the determination of specific heat: momentary temperature compensation is extrapolated through prolongation of both straight graph branches. If a parallel line to the temperature axis is drawn in such a way that two equal surfaces A_1 and A_2 are formed, the difference of the ordinates of the points of intersection of the two straight lines is the temperature difference T sought after.

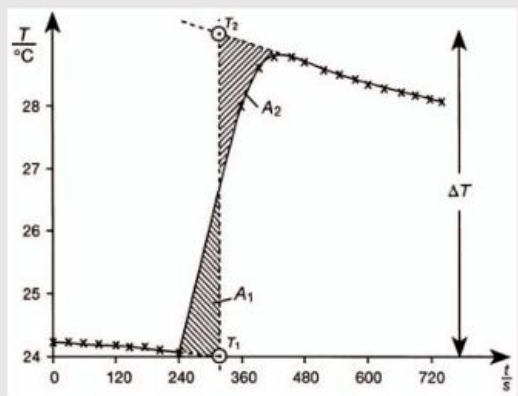


Fig 3: Temperature-time diagram for a measurement example.

The cylinder is cooled mainly due to convection. In this case we can use the Newton`s law of cooling as a good approximation: "The rate of heat loss of a body is proportional to the difference in temperatures between the body and its surroundings". In this approximation the temperature difference in Fig. 3 corresponds to the heat loss rate (first derivative of total heat amount). The integral of temperature difference thus equals to the total amount of lost heat, which graphically corresponds to the area confined under the temperature curve in Fig. 3. There are two curves in Fig. 3. The first one (real experiment) corresponds to the mechanical work slowly transferred into heat. The second curve (imaginary experiment) shows instantaneous transfer. In both experiments the cylinder has the same amount of heat in the beginning of the experiment and after heating due to friction. Mechanical work is the same for both curves. Thus the amount of lost heat (area under curves) due to convection must be the same for both curves. This is possible only if , $A_1 = A_2$ which uniquely defines the position of the vertical line (instantaneous process) in Fig. 3.

Exponential cooling process can be approximated with straight lines for short time intervals.

In the case of a measurement example where the residual load on the dynamometer is F_D= 3 N, work after 200 rotations of the friction cylinder amounts to:

ΔW=1301 J

According to equation (3), the resulting temperature increase of the friction cylinder, ΔT = 5.1 K, produces the following amount of thermal energy:

ΔQ=1296 J

$$\frac{\Delta W}{\Delta Q} = 1 \tag{6}$$

If the mechanical heat equivalent is calculated, the quotient (1.003) amounts to 1 within the limits of measurement error. As heat is a form of energy, the total amount of mechanical energy must be transformed into an equal amount of heat energy according to the law of conservation of energy. In fact, the mechanical equivalent of heat must thus amount to:

Similar experiments, in which electric energy is converted to thermal energy, show that the corresponding electric equivalent of heat $\Delta W_{el}/Q$ also amounts to 1. These results are comprehensively summarized in the first law of thermodynamics.

The change of internal energy ΔU of a system is equal to the amount of heat energy ΔQ and to the amount of work ΔW supplied from outside, independently of the supply procedure:

$$\Delta U = \Delta Q + \Delta W \tag{7}$$

If, inversely, equation 6 is assumed ($\Delta Q = \Delta W$), the specific thermal capacity can be determined with this equipment. If equations 3 and 5 are transformed according to the thermal capacity of the friction cylinder, the following result is obtained:

$$C = -8\,\frac{J}{gK} \tag{8}$$

The specific thermal capacity, defined as the quotient between the thermal capacity of a body C and its mass:

$$c = C/m \tag{9}$$

is a material constant, which at normal ambient room temperatures and atmospheric pressure depends very little on absolute temperature. It is to be noted from equation 9 that the thermal capacity of a homogeneous material is strictly proportional to its mass.

An example measurement with a rest load F = 4 N on the dynamometer, 200 turns of the crank, and a temperature increase of ΔT = 2.5 K yielded a thermal capacity of

$$C = 501\,J/gK$$

for a brass cylinder of 1280 g. (that is actually about twice the thermal capacity of a cylinder of half the weight (247 J / gK)). The specific thermal capacity calculated with equation 9 is:

$$C_{brass} = 501\,J/gK$$

The measurement of the aluminium cylinder yielded the following value:

$$C_{al} = 0.870\, J/gK$$

as compared to the theoretical value $C_{al} = 0.902\, J/gK$ for the specific thermal capacity of aluminium at room temperature.

Equipment

Friction Cylinder CuZn (Brass), m=1.28 kg, Friction Cylinder Aluminum, m=0.39 kg, Digital stopwatch, Weight 5 kg, Dynamometer, Thermocouple, Multimeter, Thermometer, Heat transfer paste.

Set-up and Prodecure:

Part I

1. First, secure the friction cylinder and operating lever by tightening them, and clamp the base plate tightly to the table with a screw. An error at this stage may cause the base plate to tilt during operation, which could lead to the thermometer breaking. Turn the friction band attached to the dynamometer around the friction cylinder 2.5 times (so that the dynamometer loosens when the lever is turned clockwise). Add a 5 kg weight to the lower end of the friction band.

2. To read the temperature, carefully insert the thermometer into the hole in the center of the friction cylinder. The hole in the cylinder and the thermometer must be carefully aligned; this is important to prevent the thermometer from breaking while the cylinder is rotating. To improve thermal contact, the hole in the cylinder is filled with a heat-conducting material.

3. At the start of the measurement, note the initial temperature reading and, by rotating the operating lever quickly and steadily 200 times, record the temperature in Table I every 30 seconds for 4 minutes. At the end of 4 minutes, when the temperature reaches its maximum, read the F_D force from the dynamometer. After the cycle is complete, record the temperatures in Table I every 30 seconds.

4. Draw the temperature graph as a function of time as shown in Figure 2. Read the temperature change value $\Delta T = T_2 - T_1$ from this graph.

5. 5. Using equations (3) and (5), calculate ΔQ. Using the F_D value read in step 3, calculate

ΔW. Find the ratio $\Delta W/\Delta Q$.

$$\frac{\Delta W}{\Delta Q} =$$

Table I:

t (s)	T (^{0}C)	t (s)	T (^{0}C)	t (s)	T (^{0}C)	t (s)	T (^{0}C)	t (s)	T (^{0}C)
0		150		300		450		600	
30		180		330		480		630	
60		210		360		510		660	
90		240		390		540		690	
120		270		420		570		720	

Part II

Perform the experiment similarly and determine the heat capacity of the aluminum cylinder. Before starting the experiment, the friction cylinder and friction belt should be cleaned with a dry cloth to remove metallic wear dust. When using an aluminum cylinder, the friction force should not exceed 10 N to prevent heavy wear and staining. For this, use a dynamometer with a 1 kg weight and 10 N accuracy.

1. Repeat the steps in Part I of the experiment for the aluminum cylinder. Record the measurements in Table II.
2. Determine the approximate value of $\Delta T = T2 - T1$ from Table II.
3. Calculate the heat capacity (C) of the cylinder using Equation (8), with the number of rotations of the arm n, the temperature increase ΔT and the force F_D read from the dynamometer.
4. Calculate the specific heat capacity (c) of the cylinder using Equation (9).

Table I

t (s)	T (^{0}C)	t (s)	T (^{0}C)	t (s)	T (^{0}C)	t (s)	T (^{0}C)	t (s)	T (^{0}C)
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30		180		330		480		630	
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EXPERIMENT-RELATED QUESTIONS

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2. . Debye's theory can accurately explain heat capacity at low temperatures. But is Debye's theory valid at high temperatures? Investigate.
3. Another theory proposed to explain heat capacity in solids is known as Einstein's theory. Recreate heat capacity using Einstein's theory. Draw the heat capacity expression you obtained as a function of temperature and compare it with the results of Debye's theory.
4. . What is latent heat? Is there a relationship between latent heat and heat capacity? Investigate.
5. In which of the reversible and irreversible processes is the heat change greater?
6. What are the laws of thermodynamics? Explain.
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REFERENCES

1. PHYWE series of publications, Laboratory Experiments

04- THERMAL AND ELECTRICAL CONDUCTIVITY OF METALS

PRINCIPLE

The thermal conductivity of copper and aluminium is determined in a constant temperature gradient from the calorimetrically measured heat flow.

The electrical conductivity of copper and aluminium is determined and the Wiedmann-Franz law is tested.

TASKS

1. Determine the heat capacity of the calorimeter in a mixture experiment as a preliminary test. Measure the calefaction of water at a temperature of 0 °C in a calorimeter due to the action of the ambient temperature as a function of time.
2. To begin with, establish a constant temperature gradient in a metal rod with the use of two heat reservoirs (boiling water and ice water). After removing the pieces of ice, measure the calefaction of the cold water as a function of time.
3. Determine the electrical conductivity of copper and aluminium by recording a current-voltage characteristic line.
4. Test of the Wiedmann-Franz law.

THEORY

1. **Thermal Conductivity of Metals**

In insulating and non-metallic solids, heat is conducted entirely by phonons, meaning heat transfer can be explained through lattice vibrations. In metals, however, both electrons and phonons contribute to heat conduction. In many metals, the contribution of electrons is 10^2 orders of magnitude greater than the contribution of phonons and is therefore negligible. Therefore, in the mid-temperature region, thermal conductivity in metals is caused by the heat carried by the degrees of freedom of electrons, which can be considered as an electron gas. The Drude model and the free electron model successfully explain thermal conductivity in metals and the Wiedeman-Franz law in this context.

Below is a brief overview of the theoretical information required for the experiment. Thermal flux density is given by Fourier's law:

$$\vec{J} = -\lambda \overrightarrow{\nabla T}$$

If a temperature difference exists between different locations of a body, heat conduction occurs. In this experiment there is a one-dimensional temperature gradient along a rod. The quantity of heat dQ transported with time dt is a function of the cross-sectional area a and the temperature gradient δT/ δx perpendicular to the surface.

$$\frac{dQ}{dt} = -\lambda A . \frac{\partial T}{\partial x} \tag{1}$$

λ is the heat conductivity of the substance.

The temperature distribution in a body is generally a function of location and time and is in accordance with the Boltzmann transport equation

$$\frac{\partial T}{\partial t} = \frac{\lambda}{p.c} . \frac{\partial^2}{\partial^2} \tag{2}$$

After a time, a steady state

$$\frac{\partial T}{\partial x} = 0 \tag{3}$$

is achieved if the two ends of the metal rod having a length l are maintained at constant temperatures $T_1$and T_2, respectively, by two heat reservoirs.

Substituting equation (3) in equation (2), the following equation is obtained:

$$T(x) = \frac{T_2 - T_1}{l} . \mathrm{x} + T_1 \tag{4}$$

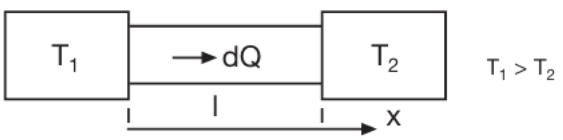
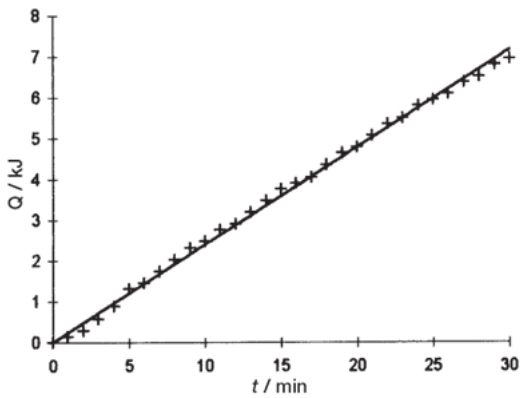


Fig. 3. Diagram: Heat of surroundings over time.



1. The heat capacity of the calorimeter is obtained from results of the mixing experiment and the following formula:

$$C_k = c_w . m_w \frac{\vartheta_w - \vartheta_M}{\vartheta_w - \vartheta_R}$$

c_w= Specific heat capacity of water

m_w = mass of the water

ϑ_w = temperature of the hot water

ϑ_m = temperature of the mixture

ϑ_R is the temperature of the room

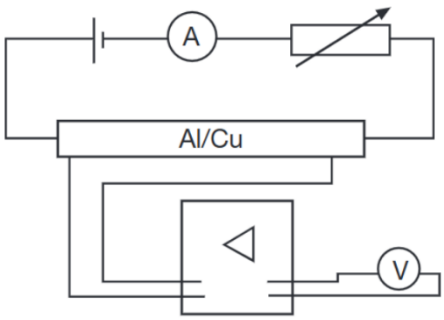
The measurement supplies a value of approximately 78 J/K ± 25%. The large variations in the results are a result of the manner in which the experiment is performed and of the experimental set-up. The addition of heat from the surroundings is calculated from the temperature increase (T of the cold water in the calorimeter.

$$\Delta Q = (c_w . m_w + C). \Delta T \tag{6}$$

where $\Delta T = T - T_0$

T_0= Temperature at time t=0

2. The heat energy supplied to the lower calorimeter can be calculated using Equation (6). The values and the change in the temperature difference on the metal rod are plotted as a function of time.



In the diagram illustrating the temperature difference, one can see that the temperature essentially remains constant. Consequently, equation 3 can be considered as having been satisfied. In order to calculate the heat energy transported by the metal rod according to Equation 1, the ambient heat fraction must be subtracted.

$$\frac{dQ_{stab}}{dt} = \frac{dQ_{gas}}{dt} = -\frac{dQ_{Umg}}{dt} \tag{7}$$

dQ/ dt for the ambient heat can be calculated from the slope of the graph in Fig. 3. dQ/dt for the entire set-up can be calculated from the slope of the graph of Q over t in Figs. 4 and 5. With the values for the length of the rod (Δx= 31.5 cm), the area (A: 4.91 · 10-4m^2) and the

averaged temperature on the metal rod, the heat conduction number can be calculated using Equation (1). The following result as the average values:

$$\lambda_{Al} = 254 \frac{W}{Km}$$

$$\lambda_{Cu} = 447 \frac{W}{Km}$$

The literature values are:

$$\lambda_{Al} = 220 \frac{W}{Km}$$

$$\lambda_{Cu} = 384 \frac{W}{Km}$$

Fig. 4a: ΔT as function of time for aluminium.

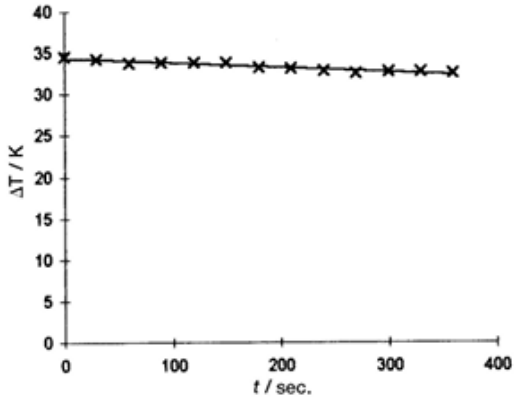
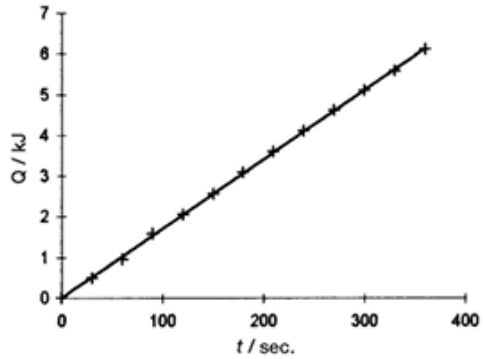


Fig. 4b: Q as a function of time for aluminium.



3. At room temperature the conduction electrons in metal have a much greater mean free path than the phonons. For this reason heat conduction in metal is primarily due to the electrons. The resulting correlation between the thermal conductivity and the electrical conductivity is established by the Wiedmann-Franz law:

$$\frac{\lambda}{\sigma} = LT \tag{8}$$

The Lorenz number L, which can be experimentally determined using Equation (8), is established by the theory of electron vapour (for temperatures above the Debye temperature) to be:

$$L = \frac{\pi^2}{3}\frac{k^2}{e^2} = 2.4.10^{-8}\ \frac{W\Omega}{K^2} \tag{9}$$

k- Universal gas constant= 1.38. $\frac{10^{-23}\ J}{K}$

E- El3mentary unit charge= 1.602. $10^{-19}\ AS$

The electrical conductivity is determined by the resistance R of the rod and its geometric dimensions (l = 0.315 m, A = 4.91 $10^{-4}m^2$).

$$\sigma = \frac{l}{A.R} \tag{10}$$

$$V = \frac{l}{\sigma A} \tag{11}$$

From Equation (8) The following values result for T= 300 K and the from the second part of the experiment.

	$R/10^{-6}\ \Omega$	$\sigma/10^{7}\ (\Omega m)^{-1}$	$L/10^{-8}\ W\Omega K^{-2}$
AL	19.6	3.27	2.5
CU	12.04	5.33	2.35

The Debye temperatures of copper and aluminium are 335 K and 419 K, respectively. Below the Debye temperature the ratio of the conductivity is smaller that given by Equation (8).

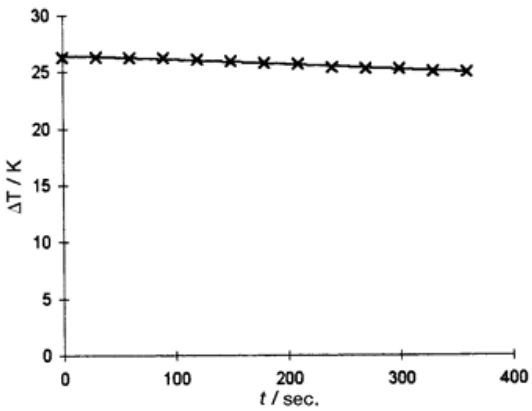


Fig. 5a: ΔT as function of time for copper.

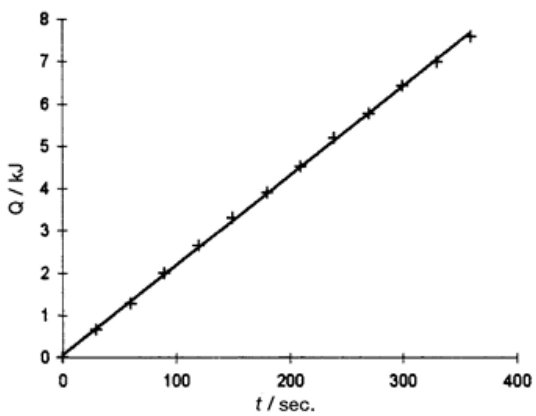


Fig. 5b: Q as function of time for copper.

Equipment for thermal measurement:

1. Calorimeter container (hot heat reservoir) 500ml
2. Calorimeter container (cold reservoir) 500ml
3. Copper/Aluminum rod (L = 0.315m, S = 4.91× 10−4m²)
4. Digital temperature gaug
5. Temperature probe (immersion type) Pt100
6. Surface temperature probes (two) Pt100

7. Heater
8. Support steel rods
9. Magnetic stirrer bar (mini)
10. Heat conductive paste
11. Connecting clamps

Equipment for electrical measurements:

1. Power Supply
2. Digital multimeters (ammeter and voltmeter)
3. Variable resistor (rheostat)
4. Amplifier
5. Connection cables
6. Copper/Aluminum rod

Set-up and Prodecure:

Part I:

Measurement of the heat capacity of the lower calorimeter

1. Weigh the calorimeter at room temperature (m).
2. Put some cold water (mass m_1) into the calorimeter container, close the container and measure its temperature (T=T_1). (To determine the mass of the water, measure the mass of the container filled with water and subtract the mass of the empty container).
3. Add some hot water (mass m_2) with a known temperature (T=T_2) to the calorimeter container.
4. Measure the temperature (T=T) when the mixture reaches thermal equilibrium.
5. Calculate the heat capacity of the calorimeter container using the following formula.

$$C_k = c_w.\frac{[m_2(T_2 - T) + m_1(T_1 - T)]}{(T - T_2)}$$

6. In this experiment, the heat capacity C=78±20 J/˚K given to the manufacturer for the calorimeter vessel can be used.

Part II:

Determining the Effect of the Environment

To find the thermal conductivity coefficient, it is necessary to find the heat flowing through the rod to the cold reservoir per unit time in steady state.

However, heat also flows from the environment to the cold reservoir during the experiment. To calculate the heat flowing from the environment to the cold reservoir, Table I is created

by following the procedure below.

1. Measure the mass of the calorimeter container.
2. Put some water (200-300 ml) into the calorimeter container.
3. Place the magnetic bar in the calorimeter container.
4. Put ice in a small cloth bag and immerse it in the water in the calorimeter container.
5. Place the calorimeter container on the magnetic stirrer and start it; this will cause the magnetic bar to rotate and mix the water, cooling it homogeneously.
6. Measure the temperature of the water-ice mixture with an immersion type temperature probe.
7. Wait until the temperature of the water-ice mixture is approximately 0°C.
8. When the temperature is approximately 0°C, remove the ice from the water in the calorimeter container and record the temperature at that moment as T0. From this point, measure the water temperature at one-minute intervals for 20 minutes and record it in Table 1.
9. After the measurements are complete, remove the magnetic bar from the water and measure the mass of the calorimeter container filled with water.
10. Subtract the mass of the container you measured earlier from this mass and record the mass of the water (m_{w1}).
11. $\Delta Q = (c_w . m_w + C).\Delta T \ , \quad \Delta T = T - T_0 \quad , i = 1, \dots .20$
12. $\Delta Q_{env.} - t$
13. $\Delta T - t$
14. Slope of $\Delta Q_{env.} - t$ gives the heat flowing from the environment to the calorimeter container per unit time $\frac{d\Delta Q_{env.}}{dt}$.

Table I:

$m_{w1} =$ gr $T_0 =$ ^{0}K			
t (mins)	T (^{0}K)	ΔT(rod)(^{0}K)	ΔQ_{total}(J)
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			

Part III:

Determination of the thermal conductivity

1. Perform the experimental set-up according to Fig.

2. Weigh the empty, lower calorimeter.

3. Insert the insulated end of the metal rod into the upper calorimeter vessel. To improve the heat transfer, cover the end of the metal rod with heat-conduction paste.

4. Attach the metal rod to the support stand in such a manner that the lower calorimeter can be withdrawn from beneath it.

5. The height of the lower calorimeter can be changed with the aid of the supporting block. When doing so, care must be taken to ensure that the non-insulated end of the rod remains completely immersed in the cold water during the experiment.

6. The surface temperature probe must be positioned as close to the rod as possible.

7. The outermost indentations on the rod (separation: 31.5 cm) are used to measure the temperature difference in the rod. To improve the heat transfer between the rod and the surface probe, use heat-conduction paste.

8. Using an immersion heater, bring the water in the upper calorimeter to a boil, and keep it at this temperature.

9. Ensure that the upper calorimeter is well filled to avoid a drop in temperature due to contingent refilling with water.

10. Keep the water in the lower calorimeter at 0°C with the help of ice (in a gauze pouch).

11. Observe the ΔT temperature difference between two separate points on the probe (the point closest to the upper calorimeter container and the point closest to the lower calorimeter container) using surface temperature probes and a digital thermometer.

12. Measurements can begin when a constant temperature gradient (i.e., the ΔT temperature difference between the two points remains constant) is achieved (this may take 15-20 minutes to achieve).

13. When ΔT is constant, remove the ice from the cold water, record the temperature you read from the digital thermometer with the submerged probe in the lower calorimeter as T_0.

14. Simultaneously record the ΔT temperature difference from the digital thermometer.

15. For 20 minutes, at one-minute intervals, read the water temperature in the lower calorimeter container and the ΔT temperature difference on the probe from the digital thermometer and record them in Table II.

16. The temperature difference data (ΔT) obtained on the rod will be used to determine the value of $\frac{dT}{dx} = \frac{T_2 - T_1}{x_2 - x_1}$

17. When the measurements are finished, measure the mass of the lower calorimeter container together with the water inside and subtract the mass of the empty container to find the mass of the water (m_{w2}) and write it in Table 2.

18. Draw the graph of ΔT (temperature differences measured on the rod)$-t$ and fit it to a line. From this, the steady state can be seen following eq.

$$\frac{dQ_{\Delta x}}{dt} = \rho S \Delta x c_m \frac{\partial T(x,t)}{\partial t}$$

19. Calculate the heat flowing from the surroundings and through the rod to the lower calorimeter vessel using Eq. $\Delta Q = (c_w . m_w + C).\Delta T$, $\Delta T = T - T_0$ $, i = 1, \ldots .20$ for each temperature value (ΔQtpm) and write it in Table II.

20. Draw the graph of $\Delta Q_{tot.} - t$. The slope of this graph gives the heat flowing from the surroundings and through the rod to the lower calorimeter vessel per unit time.

21. Calculate the thermal conductivity coefficient ε (for copper/aluminum) using the following equation and compare it with the value in the literature:

22. $$\frac{dQ}{dt} = \frac{dQ_{tot.}}{dt} - \frac{dQ_{env.}}{dt} = -\lambda A . \frac{\partial T}{\partial x}$$

23. For dT, calculate the arithmetic mean of the temperature differences ΔT measured across the rod, and for $dx = x_2 - x_1$, calculate the distance between the two surface probes on the rod.

Table II:

m_{w2} = gr		T_0 = ^{0}K		
t(min)	T (^{0}K)	ΔT(rod)(^{0}K)	ΔQ_{tot}(J)	$\Delta T = T - T_0$(^{0}K)
1				
2				

3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
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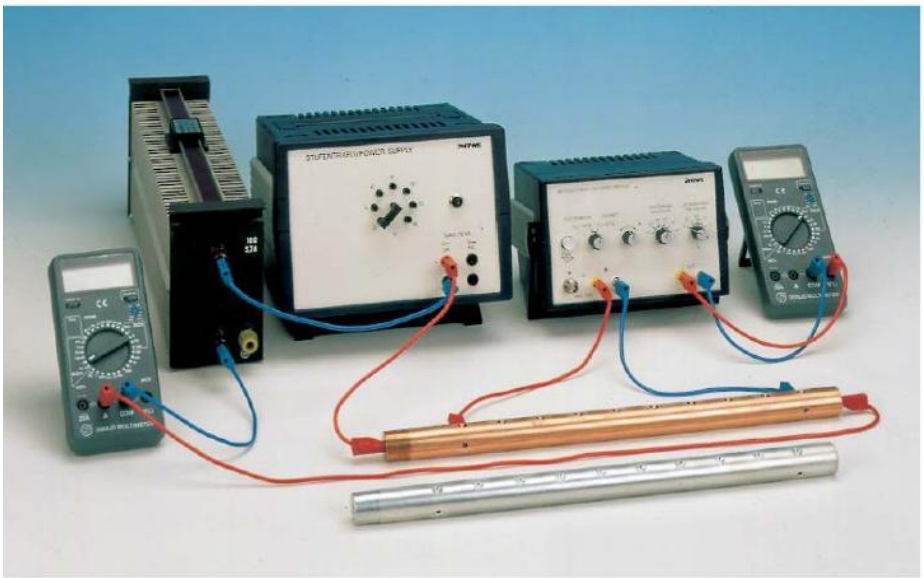
Part IV:

Measurement of the electrical conductivity.

1. Perform the experimental set-up according to the circuit diagram in Fig. 2 (set-up in accordance with a 4-conductor measuring method).



2. Set the voltage on the variable transformer to 6 V.
3. The amplifier must calibrated to 0 in a voltage-free state to avoid a collapse of the output voltage.
4. Select the amplifier settings as follows:
   Input: Low Drift
   Amplification: 10^4
   Time constant: 0
5. Set the rheostat to its maximum value and slowly decrease the value during the experiment.
6. Read and note the values for current and voltage
7. The resistance, and thus the electrical conductivity, can be determined from the measured values.
8. Determine the resistance and electrical conductivity values from the experimental data using Eqs. (10) and (11) respectively and write them in Table III. The length of the Al/Cu rod is L=0.315m and the cross-sectional area is $A=4.01\times10^{-4}m^2$.

Part V:

Testing the Wiedemann-Franz Law

Write the thermal and electrical conductivity values you found in Eq. (10) and determine the Lorentz constant L for Al/Cu metal at room temperature (300˚K) and compare it with the value given in Table IV.

Table III:

V (Volt)	I (Amp.)	R (Ω)	σ ($\Omega^{-1}m^{-1}$)

Table IV:

	$R[10^{-6}\Omega]$	$\sigma[10^{7}(\Omega m)^{-1}]$	$L[10^{-8}W\Omega K^{-2}]$
Al	19.60	3.27	2.50
Cu	12.04	5.33	2.35

REFERENCES

1. PHYWE series of publications, Laboratory Experiments